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Question 1

Not answered

Marked out of 1

An epidemic of wide geographical distribution is known as

Select one:

- ☐ Hyperepidemic disease
- ☐ Endemic disease
- ☐ Pandemic disease
- ☐ Hyperendemic disease
- ☐ Sporadic disease

Your answer is incorrect.

The usual prevalence of a disease in a community is called the baseline prevalence. If the baseline is a low to moderate level and continues without significant fluctuation, the disease is said to be endemic in the community. A persistent but high level of baseline is called hyper endemic occurrence. Most psychiatric disorders are low level endemic disorders (with exceptions). An irregular pattern of significant fluctuation from baseline is called sporadic occurrence. When the occurrence within an area clearly exceeds the expected level in a given time period, it is called an epidemic. An epidemic can be slow and steady rising epidemic as in the case of multifactorial chronic diseases or acute outbreaks as in infectious diseases. A pandemic is an epidemic in wide geographical proportion g. HIV pandemic.

The correct answer is: Pandemic disease

Question 2

Not answered

Marked out of 1

Most common level of prevention currently employed in psychiatric practice is

Select one:

- ☐ Protective
- ☐ Primary
- ☐ Secondary
- ☐ Prodromal
- ☐ Tertiary

Your answer is incorrect.

In current psychiatric practice, our knowledge of risk factors and causal insults is limited. Hence most prevention in psychiatric practice is of tertiary nature. Early intervention program in psychosis is an example of secondary prevention. Public health initiatives such as eradication of poverty, maintaining healthy diet etc could prevent certain, at least milder forms of mental illnesses - these could be termed as primary prevention strategies. Interventions aimed at high risk groups are usually secondary preventions. Rehabilitation is a tertiary prevention (reducing disabilities).

The correct answer is: Tertiary

Question 3

Not answered

Marked out of 1

The incidence of schizophrenia is three times higher in migrants, but this drops to 1.8 times higher than natives when point prevalence rates are considered. This can be explained by

Select one:

- ☐ Better access to care for natives
- ☐ Shorter duration of illness in migrants
- ☐ Higher death rate in natives
- ☐ Better treatment response in natives
- ☐ Longer illness duration in migrants

Your answer is incorrect.

For illnesses which are significantly chronic (e.g. schizophrenia), prevalence will be higher compared to those illnesses which are acute and short-lived (g. influenza), even if the incidence rates are comparable. Hence a simple expression $\text{Prevalence} = \text{incidence} \times \text{duration of illness}$ explains the relationship between incidence and prevalence. One explanation for disparity observed in migrants to natives ratio of incidence and prevalence may relate to the relatively shorter duration of psychoses in immigrants. This is hypothetical and not a widely established explanation.

The correct answer is: Shorter duration of illness in migrants

Question 4

Not answered

Marked out of 1

The annual incidence of injected cocaine use is estimated to be 20/100,000. The annual mortality rate is 4/100,000; if point prevalence is 80/100,000 what is the calculated duration of cocaine use?

Select one:

- ☐ 8 years
- ☐ 5 years
- ☐ 40 years
- ☐ 1 year
- ☐ 4 years

Your answer is incorrect.

Prevalence = incidence X duration of illness explains the relationship between incidence and prevalence. If we discard the very small mortality rate of 4/100,000, and apply the above formula, $80 = 20 \times (?)$, we can arrive at Duration = 4 years.

The correct answer is: 4 years



Question 5

Not answered

Marked out of 1

Which of the following is expressed as number of new cases per person-year of observation?

Select one:

- ☐ Incidence rate
- ☐ Lifetime morbid risk
- ☐ Point prevalence
- ☐ Cumulative incidence
- ☐ Period prevalence

Your answer is incorrect.

Incidence (of a disease in year 2008 at an area A) = Number of newly diagnosed cases in year 2008 / Mid 2008 population at area A. Incidence is a ratio measured against time. One month incidence will be invariably lower than one year incidence. The most common measurement of incidence using the above formula is also known as cumulative incidence. This is valid in cohort studies where attrition is low, but not so useful when attrition is significant. This is because dropouts are excluded from being counted as new cases, while the denominator population usually takes them into account. To avoid this problem, incidence density (also known as incidence rate) is used. It expresses number of new cases per person-year of observation. The unit of the incidence rate is rather bizarre, given by person-time units. Hence incidence rates are often used in reporting results that compare two groups incidence rates - providing a ratio with no units.

The correct answer is: Incidence rate

Question 6

Not answered

Marked out of 1

Which of the following rates applies to the whole of a population and not a selected subgroup of the population?

Select one:

- ☐ none of the above
- ☐ age specific mortality
- ☐ crude mortality rate
- ☐ disease specific mortality
- ☐ cause specific mortality rate

Your answer is incorrect.

A crude rate is any rate applicable to the whole population, e.g., crude birth rate, crude mortality rate, etc. A specific rate is any rate applicable to a subgroup of a population e.g. age-specific mortality, disease-specific mortality, cause-specific mortality rates et

The correct answer is: crude mortality rate



Question 7

Not answered

Marked out of 1

There are 65 suicides in a population of 1300 patients with schizophrenia. The rate 65/1300 refers to which one of the following?

Select one:

- ☐ Standardised mortality rate
- ☐ Cause specific mortality rate
- ☐ Proportional mortality rate
- ☐ Case fatality rate
- ☐ Crude mortality rate

Your answer is incorrect.

Cause-specific mortality rate: g. alcohol specific mortality. This refers to ratio between number of deaths due to alcohol in a population and total population size. Case-fatality rate: Ratio between number of deaths due to a specific disease and number of persons affected by the disease in a population. It is a measure of the fatal severity of the disease studies. For example, 15 patients out of 100 with anorexia will die due to its complications. Proportionate mortality rate: It is a measure of contribution of a disease to societal mortality burden. It is given by the ratio between deaths due to a specific cause and total number of deaths in a population. In this question, out of 1300 persons in a population with schizophrenia, death due to suicide (cause) is 65. Hence 65/1300 is cause specific mortality rate. [Note that these 65 did not die because of schizophrenia - the cause of their death is suicide]. In a similar fashion one can expand this to stroke-specific mortality in schizophrenia, Myocardial infarction specific rate etc.

The correct answer is: Cause specific mortality rate

Question 8

Not answered

Marked out of 1

The probability of developing a disorder anytime throughout the life course of a birth cohort is called as

Select one:

- ☐ Period prevalence
- ☐ Life time morbid risk
- ☐ Cumulative incidence
- ☐ Life expectancy
- ☐ Lifetime prevalence

Your answer is incorrect.

Lifetime prevalence is "the proportion of individuals in the population who have ever manifested a disorder, who are alive on a given day. Lifetime prevalence needs to be clearly distinguished from lifetime morbid risk (LMR). LMR is the probability of a person developing the disorder during entire period (often a specified period, defined by the life expectancy of the population studied) of their life. LMR includes the entire lifetime of a birth cohort both past and future, and includes those deceased at the time of the survey. For low-incidence disorders such as schizophrenia, summation of age-specific incidence rates gives approximate LMR" (Saha et al PLoSMed 2004).

The correct answer is: Life time morbid risk

Question 9

Not answered

Marked out of 1

Which of the following data is published after adjusting for the age of the population studied?

Select one:

- ☐ Perinatal mortality rate
- ☐ Case fatality rate
- ☐ Infant death rate
- ☐ Standardised mortality ratio
- ☐ Crude mortality rate

Your answer is incorrect.

Infant death rate the observed sample consists only of children age less than 1 year. This is not same as making an adjustment after observation. The correct answer for this question is standardized mortality wherein adjustments are made according to the distribution of age and other variables after observation. (Ref: Naing, N. <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3406211/>)

The correct answer is: Standardised mortality ratio

Question 10

Not answered

Marked out of 1

Which of the following is often used to display the age and sex distribution of a population?

Select one:

- ☐ Forest plot
- ☐ Chi Square 2X2 table
- ☐ Stem and Leaf plot
- ☐ Correlation coefficient
- ☐ Bar chart

Your answer is incorrect.

Bar charts are easy to produce and interpret.

The correct answer is: Bar chart



Question 11

Not answered

Marked out of 1

Which of the following factors is least likely to influence the interpretation of the comparison between two crude mortality rates?

Select one:

- ☐ Socio-economic stratification
- ☐ Ethnic composition
- ☐ Age distribution
- ☐ Gender distribution
- ☐ Size of the populations

Your answer is incorrect.

As population size enters as a denominator, it will not influence the comparisons. But two geographical regions can vary in the other listed factors that can make comparisons of crude mortality rates meaningless.

The correct answer is: Size of the populations



Question 12

Not answered

Marked out of 1

The age and sex structure of a population are best illustrated as a

Select one:

- ☐ Funnel plot
- ☐ Forest plot
- ☐ Life table
- ☐ Population pyramid
- ☐ Pie chart

Your answer is incorrect.

A population pyramid, also called an age structure pyramid, is a graphical illustration that shows the distribution of various age groups in a human population which normally forms the shape of a pyramid. It is often considered to be the most effective way of graphically depicting the age and sex distribution of a population, partly because of the very clear images that the pyramids present.

The correct answer is: Population pyramid

Question 13

Not answered

Marked out of 1

When a cohort study was initiated in a hospital catchment area, it was determined that at the outset 200 of 1200 'at risk people' in the cohort already had the illness under study. Over the next year, 150 students contracted the illness for the first time. What was the incidence rate?

Select one:

- ☐ 350/1000
- ☐ 150/1000
- ☐ 350/1200
- ☐ 150/1200
- ☐ 200/1200

Your answer is incorrect.

At risk population becomes 1000 now.

The correct answer is: 150/1000



Question 14

Not answered

Marked out of 1

The phenomenon of relative deprivation is explained by

Select one:

- ☐ Jarmen index
- ☐ Brian Index
- ☐ Haschinski index
- ☐ Barthel index
- ☐ Jasper Index

Your answer is incorrect.

A scoring system developed by the British general practitioner Brian Jarman for the level of social deprivation in a community, using census data on percentages of old people living alone, single-parent families, children younger than 5 years of age, unskilled and unemployed persons, ethnic minorities, overcrowded dwellings, changes of address in previous year, etc. Although a valid indicator, it is not generally accepted outside the United Kingdom.

The correct answer is: Jarmen index

Question 15

Not answered

Marked out of 1

Which of the following epidemiological pattern is true for a rare but highly fatal disease?

Select one:

- ☐ Low incidence and high prevalence
- ☐ Low incidence and prevalence
- ☐ High incidence and low prevalence
- ☐ High incidence and prevalence
- ☐ Zero prevalence

Your answer is incorrect.

As the disease is fatal most (not all) patient will die sooner rather later, leaving a low prevalence rate. As it is a rare condition, the incidence will be low.

The correct answer is: Low incidence and prevalence



Question 16

Not answered

Marked out of 1

In 1999, 50 cases of CJD had been linked to French-derived hGH, from a cohort of 1500 patients. Of these 12 died at the time of the survey. Calculate the attack rate for CJD transmission in this cohort.

Select one:

- ☐ 38/1450
- ☐ 50/1500
- ☐ 38/1500
- ☐ Insufficient data to calculate
- ☐ 50/1450

Your answer is incorrect.

Attack rate is given by number of cases that develop divided by number exposed in a cohort (at risk). It is simply an incidence rate following an acute exposure.

The correct answer is: 50/1500



Question 17

Not answered

Marked out of 1

The following descriptions of levels of prevention are correct except

Select one:

- ☐ Continuous treatment to enable chronic patients to attain best life quality is secondary prevention
- ☐ Immunization is considered primary prevention
- ☐ Primary prevention is the most effective step in all levels of prevention
- ☐ Detection and treatment of disease belong to the secondary prevention
- ☐ Rehabilitation is part of tertiary prevention

Your answer is incorrect.

Continuous treatment to enable chronic patients to attain best potential of functioning is an important part of tertiary, not secondary, prevention.

The correct answer is: Continuous treatment to enable chronic patients to attain best life quality is secondary prevention



Question 18

Not answered

Marked out of 1

A researcher studies data from Office of National Statistics in UK and finds out that at areas where number of consultant psychiatrists per catchment population is high, the suicide rates are increased. He suggests that mental health services may be driving people to kill themselves unknowingly. This is correctly termed as

Select one:

- ☐ Measurement Bias
- ☐ Selection Bias
- ☐ Sampling error
- ☐ Ecological fallacy
- ☐ Type 2 error

Your answer is incorrect.

The Ecological Fallacy is a situation that can occur when a researcher or analyst makes an inference about an individual based on aggregate data for a group.

The correct answer is: Ecological fallacy



Question 19

Not answered

Marked out of 1

According to the pathways of care model, illness behaviour is a strong determinant at which of the following filters between disease and secondary psychiatric care?

Select one:

- ☐ Third stage - Acceptance of referral
- ☐ Second stage - GP recognition
- ☐ Fifth stage - availability of hospital beds
- ☐ Fourth stage - community psychiatric services
- ☐ First stage - self recognition

Your answer is incorrect.

Illness behaviour varies widely even among people with same degree and quality of an illness. This could explain the variation in self-recognition and help seeking.

The correct answer is: First stage - self recognition



Question **20**

Not answered

Marked out of 1

A psychiatrist encourages his team members to undergo influenza vaccination. He says 'If not anything it will prevent some post-influenza depression and reduce our case load'. Which of the following prevention is he referring to?

Select one:

- ☐ Secondary prevention
- ☐ Prodromal prevention
- ☐ Primary prevention
- ☐ Rehabilitation
- ☐ Tertiary prevention

Your answer is incorrect.

Primary prevention refers to interventions such as vaccination or public education that can prevent or avert an illness/epidemic.

The correct answer is: Primary prevention



Question **21**

Not answered

Marked out of 1

According to the pathways of care model, the first filter between disease and healthcare delivery is

Select one:

- ☐ Recognition of the illness by the specialist team
- ☐ Availability of crisis community services
- ☐ Diagnosing ability of General practitioner
- ☐ Availability of outpatient secondary care
- ☐ Recognition of symptoms by the patient

Your answer is incorrect.

Pathways of care or filter model was proposed by Goldberg. It explains the difference in prevalence rates across the various settings from community to tertiary centres.

The correct answer is: Recognition of symptoms by the patient



Question **22**

Not answered

Marked out of 1

Which of the following includes stillbirths and deaths in the first week compared against the total number of live births in a population?

Select one:

- ☐ infant mortality rate
- ☐ perinatal mortality rate
- ☐ crude mortality rate
- ☐ proportionate mortality rate
- ☐ case fatality rate

Your answer is incorrect.

The World Health Organization defines perinatal mortality as the "number of stillbirths and deaths in the first week of life per 1,000 live births"

The correct answer is: perinatal mortality rate



Question **23**

Not answered

Marked out of 1

The term used to describe the probability of outcome occurring at a particular time is

Select one:

- ☐ Heterogeneity
- ☐ Sampling
- ☐ Sensitivity analysis
- ☐ Likelihood ratio
- ☐ Cumulative incidence

Your answer is incorrect.

Cumulative incidence is the incidence calculated using a period of time during which all of the individuals in the population are considered to be at risk for the outcome. It is also called the incidence proportion.

The correct answer is: Cumulative incidence



Question **24**

Not answered

Marked out of 1

An age-specific mortality rate is

Select one:

- ☐ Used to give an accurate measure of standard mortality rate
- ☐ Used to compare mortality between different age groups
- ☐ Used to compare mortality between same age groups
- ☐ Used to compare incidences between 2 different conditions
- ☐ Used to predict mortality rate in a given population at specific point of time

Your answer is incorrect.

Crude mortality rates are calculated by dividing the total number of deaths in a given time period by the total number of persons in the population. The problem with crude rates is that any comparison between two crude rates is confounded by variations in the age structure within the two compared populations. Age specific rates are better suited for comparisons.

The correct answer is: Used to compare mortality between different age groups

Question **25**

Not answered

Marked out of 1

Number of deaths of live infants (one year of age or younger) per 1000 live births is called

Select one:

- ☐ Postnatal mortality rate
- ☐ Neonatal mortality rate
- ☐ Perinatal mortality rate
- ☐ Stillbirth rate
- ☐ Infant mortality rate

Your answer is incorrect.

Deaths under 1 year of age expressed as a proportion of live births is an important public health index used by the UNICEF and the WHO. This is called as infant mortality rate.

The correct answer is: Infant mortality rate



Question **26**

Not answered

Marked out of 1

What will happen to incidence and prevalence rates of schizophrenia if a new wonder drug can cure schizophrenia after the first episode of psychosis?

Select one:

- ☐ Incidence will increase, prevalence will fall
- ☐ Both Incidence and prevalence will fall
- ☐ Incidence will not be affected, prevalence will fall
- ☐ Incidence will fall, prevalence will increase
- ☐ Prevalence will not be affected, incidence will fall

Your answer is incorrect.

Prevalence = incidence x duration. Cure effectively reduces the 'duration', and so the prevalence, while incidence continues unchanged.

The correct answer is: Incidence will not be affected, prevalence will fall



Question **27**

Not answered

Marked out of 1

If a measure is introduced to prevent the progression of an established disease in a patient, this is called

Select one:

- ☐ Late prevention
- ☐ Tertiary prevention
- ☐ Failed prevention
- ☐ Secondary prevention
- ☐ Primary prevention

Your answer is incorrect.

Rehabilitation is a type of tertiary prevention

The correct answer is: Tertiary prevention



Question **28**

Not answered

Marked out of 1

The reduction in amount of disability in diagnosed patients is associated with

Select one:

- ☐ Primary prevention
- ☐ Secondary prevention
- ☐ Quaternary prevention
- ☐ Universal prevention
- ☐ Tertiary prevention

Your answer is incorrect.

Tertiary prevention consists of prevention of disease progression and an attendant suffering after it is clinically obvious and diagnosis is well established. This act also includes the rehabilitation of disabled conditions.

The correct answer is: Tertiary prevention



Question **29**

Not answered

Marked out of 1

A 64-year-old man with schizophrenia suffers from profound amotivation and anhedonia. He is encouraged to attend day centre to reduce the disabilities accompanying these negative symptoms. Which of the following modes of prevention does the above refer to?

Select one:

- ☐ Prodromal prevention
- ☐ Secondary prevention
- ☐ Tertiary prevention
- ☐ Non specific prevention
- ☐ Primary prevention

Your answer is incorrect.

Tertiary prevention focuses on helping people manage complicated, long-term problems such as diabetes, schizophrenia etc. The goals include preventing further health deterioration and maximizing quality of life.

The correct answer is: Tertiary prevention



Question 30

Not answered

Marked out of 1

A disease, which is habitually present in a defined human population, is called

Select one:

- ☐ Endemic
- ☐ Hyperendemic
- ☐ Epidemic
- ☐ Pandemic
- ☐ Hyperepidemic

Your answer is incorrect.

The usual prevalence of a disease in a community is called as the baseline prevalence. If the baseline is a low to moderate level and continues without significant fluctuation, the disease is said to be endemic in the community. A persistent but high level of the baseline is called hyperendemic occurrence. Most psychiatric disorders are low level endemic disorders (with exceptions).

The correct answer is: Endemic



Question **31**

Not answered

Marked out of 1

Two neurotic disorders have same incidence rates in a given population. But they have different prevalence rates. The most likely explanation is

Select one:

- ☐ One disease has higher public awareness and lesser stigma than the other disease
- ☐ The two disease have marked difference in sex distribution
- ☐ One disease has better prevention strategies than the other
- ☐ One disease is highly fatal compared to the other disease
- ☐ One disease is more chronic than the other

Your answer is incorrect.

The relation between incidence and prevalence is $\text{Prevalence} = \text{Duration} \times \text{Incidence}$. Hence the above scenario could be explained by differences in chronicity of the illnesses studied. Neurotic disorders cannot be fatal in themselves, so the best choice is E, though A can be considered for a different class of disorders.

The correct answer is: One disease is more chronic than the other



Question 32

Not answered

Marked out of 1

The annual incidence rate of Alzheimer's dementia in a circumscribed population is 2 per 1000. The point prevalence is 20 per 1000. If Alzheimer's specific mortality rate is 0.5 per 1000, then the average duration of Alzheimer's dementia is

Select one:

- ☐ 10 years
- ☐ 7 years
- ☐ 1
- ☐ 15 years
- ☐ 1 year

Your answer is incorrect.

Prevalence is given by product of incidence and duration of illness. As prevalence rates already include mortality statistics ('existing cases' exclude dead ones), the mortality rate need not be used in this formula.

The correct answer is: 10 years



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